

=== GNB (1785510826) ===

```
cu_cp:
  amf:
    addr: 10.53.1.2
    port: 38412
    bind_addr: 10.53.1.3
    supported_tracking_areas:
      - tac: 7
        plmn_list:
          - plmn: "21401"
            tai_slice_support_list:
              - sst: 1
    inactivity_timer: 7200

ru_sdr:
  device_driver: zmq
  device_args: tx_port=tcp://127.0.0.1:2000,rx_port=tcp://127.0.0.1:2001,base_srate=11.52e6
  srate: 11.52
  tx_gain: 75
  rx_gain: 75

cell_cfg:
  dl_arfcn: 630000
  band: 78
  channel_bandwidth_MHz: 20
  common_scs: 15
  plmn: "21401"
  tac: 7
  ssb_scs: 15
  pdcch:
    common:
      ss0_index: 0
      coreset0_index: 6
    dedicated:
      ss2_type: common
      dci_format_0_1_and_1_1: false
  prach:
    prach_config_index: 1
  pdsch:
    mcs_table: qam64
  pusch:
    mcs_table: qam64

log:
  filename: /tmp/gnb.log
  all_level: info
  hex_max_size: 0

pcap:
  mac_enable: false
  mac_filename: /tmp/gnb_mac.pcap
  ngap_enable: false
  ngap_filename: /tmp/gnb_ngap.pcap
  e2ap_enable: true
  e2ap_du_filename: /tmp/gnb_du_e2ap.pcap
  e2ap_cu_cp_filename: /tmp/gnb_cu_cp_e2ap.pcap
  e2ap_cu_up_filename: /tmp/gnb_cu_up_e2ap.pcap

metrics:
  layers:
    enable_rlc: true
    enable_sched: true
  periodicity:
    du_report_period: 1000
```

=== UE (1785510826) ===

```
[rf]
freq_offset = 0
tx_gain = 50
rx_gain = 40
srate = 11.52e6
nof_antennas = 1

device_name = zmq
device_args = tx_port=tcp://127.0.0.1:2001,rx_port=tcp://127.0.0.1:2000,base_srate=11.52e6

[rat.eutra]
dl_eurfcn = 2850
nof_carriers = 0

[rat.nr]
bands = 78
nof_carriers = 1

max_nof_prb = 106
nof_prb = 106

[pcap]
enable = none
mac_filename = /tmp/ue_mac.pcap
mac_nr_filename = /tmp/ue_mac_nr.pcap
nas_filename = /tmp/ue_nas.pcap

[log]
all_level = info
phy_lib_level = none
all_hex_limit = 32
filename = /tmp/ue.log
file_max_size = -1

[usim]
mode = soft
algo = milenage
opc = 63BFA50EE6523365FF14C1F45F88737D
k = 223344556677889900aabbccddeeff11
imsi = 214010000000003
imei = 353490069873319

[rrc]
release = 15
ue_category = 4

[nas]
apn = srsapn
apn_protocol = ipv4

[gw]
netns = ue1
ip_devname = tun_srsue
ip_netmask = 255.255.255.0

[gui]
enable = false
```

=== DOCKER (1785510826) ===

```
#
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#
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#
```

services:

```
5gc:
  container_name: open5gs_5gc
  build:
    context: open5gs
    target: open5gs
    args:
      OS_VERSION: "22.04"
      OPEN5GS_VERSION: "v2.7.0"
  environment:
    - MONGODB_IP=127.0.0.1
    - OPEN5GS_IP=10.53.1.2
    - UE_IP_BASE=10.45.0
    - UPF_ADVERTISE_IP=10.53.1.2
    - DEBUG=false
    - SUBSCRIBER_DB=214010000000003,223344556677889900aabbccddeeff11,opc,63BFA50EE6523365FF14C1F45F88737D,8000,9,10
    - NETWORK_NAME_FULL=srsRAN
    - NETWORK_NAME_SHORT=srsRAN
    - TZ=Europe/Madrid
  privileged: true
  ports:
    - "9898:9999/tcp"
  # Uncomment port to use the 5gc from outside the docker network
  # - "38412:38412/sctp"
  # - "2152:2152/udp"
  command: 5gc -c open5gs-5gc.yml
  healthcheck:
    test: [ "CMD-SHELL", "nc -z 127.0.0.20 7777" ]
    interval: 3s
    timeout: 1s
    retries: 60
  networks:
    ran:
      ipv4_address: ${OPEN5GS_IP:-10.53.1.2}

gnb:
  container_name: srsran_gnb
  # Build info
  image: srsran/gnb
  build:
    context: ..
    dockerfile: docker/Dockerfile
    args:
      OS_VERSION: "24.04"
```

```

# privileged mode is required only for accessing usb devices
privileged: true
# Extra capabilities always required
cap_add:
  - SYS_NICE
  - CAP_SYS_PTRACE
volumes:
  # Access USB to use some SDRs
  - /dev/bus/usb/:/dev/bus/usb/
  # Sharing images between the host and the pod.
  # It's also possible to download the images inside the pod
  - /usr/share/uhd/images:/usr/share/uhd/images
  # Save logs and more into gnb-storage
  - gnb-storage:/tmp
# It creates a file/folder into /config_name inside the container
# Its content would be the value of the file used to create the config
configs:
  - gnb_config.yml
  - gnb_compose_config.yml
# Customize your desired network mode.
# current network configuration creates a private network with both containers attached
# An alternative would be `network: host`. That would expose your host network into the container. It's the easiest
networks:
  ran:
    ipv4_address: ${GNB_IP:-10.53.1.3}
  metrics:
    ipv4_address: 172.19.1.3
# Start GNB container after 5gc is up and running
depends_on:
  5gc:
    condition: service_healthy
# Command to run into the final container
command: gnb -c /gnb_config.yml -c /gnb_compose_config.yml

configs:
  gnb_config.yml:
    file: ./gnb_zmq.yml # Path to your desired config file
  gnb_compose_config.yml:
    content: |
      cu_cp:
        amf:
          addr: ${OPEN5GS_IP:-10.53.1.2}
          bind_addr: 0.0.0.0
      metrics:
        autostart_stdout_metrics: true
        enable_json: true
      remote_control:
        bind_addr: 0.0.0.0
        enabled: true

volumes:
  gnb-storage:

networks:
  ran:
    ipam:
      driver: default
      config:
        - subnet: 10.53.1.0/24
  metrics:
    ipam:
      driver: default
      config:
        - subnet: 172.19.1.0/24

```

=== NOTAS (1785510826) ===

1. PLMN (MCC=214, MNC=01) configurado como "21401" en gNB, UE (IMSI) y Core (SUBSCRIBER_DB) (Regla 1, 7a).
2. ``dl_arfcn`` 630000 seleccionado para Banda 78, cumpliendo el rango [620000, 653333] (Regla 2).
3. ``common_scs`` y ``ssb_scs`` se han establecido en 15 kHz para coherencia (Regla 3).
4. Los puertos ZMQ se han configurado de forma cruzada entre gNB y UE (Regla 4).
5. ``bind_addr`` en ``gnb_zmq.yaml`` es 10.53.1.3 y en ``gnb_compose_config.yaml`` es 0.0.0.0 (Regla 5).
6. El bloque ``ru_sdr`` se ha copiado exactamente del template, incluyendo ``srate: 11.52e6`` (Regla 6).
7. IMSI, K, OPC y el formato de SUBSCRIBER_DB se han configurado siguiendo estrictamente la Regla 7.
8. ****Nota Importante:**** El ``srate`` de ``11.52e6`` es inmutable (Regla 6). Sin embargo, para un ancho de banda de 20 MHz